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Physics-informed residual reinforcement learning via expert prior knowledge for safe and efficient autonomous merging

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ABSTRACT

Autonomous on-ramp merging is a safety-critical and highly interactive driving task that requires reliable feasibility, safety assurance, and timely maneuver completion. However, existing reinforcement learning (RL) approaches often suffer from low sample efficiency in such high-dimensional and highly dynamic environments, and pure reward-driven exploration struggles to accommodate driving safety and efficiency objectives, leading to either overly aggressive or overly conservative driving behaviors. To address this challenge, we propose PR-EPK, a physics-informed residual RL framework that integrates model-based and analytical expert knowledge into both policy learning and execution. An MPC-based prior provides dynamically feasible baseline controls, while the RL agent learns bounded residual corrections to refine expert behavior, improving exploration efficiency and convergence stability. A risk-adaptive hybrid execution mechanism further regulates residual authority online as a soft safety heuristic during execution. In addition, a decoupled dual-critic constrained learning scheme separates task-return and safety-cost estimation to mitigate objective coupling, and a progressive curriculum strategy gradually transitions from expert-guided training to autonomous policy enhancement. Simulation results across varying traffic densities demonstrate that PR-EPK achieves faster and more stable convergence, consistently higher success rates, significantly reduced collision rates and shorter average merge times compared with representative baselines. These results validate the effectiveness of the proposed framework and demonstrate that structured integration of expert knowledge within residual RL provides a practical pathway toward safe and efficient autonomous merging in highly dynamic traffic.

1. Introduction

Autonomous driving technologies have shown considerable promise in enhancing roadway capacity and reducing traffic accidents [1]. Among various driving tasks, on-ramp merging represents a particularly safety-critical scenario, where the ego vehicle (EV) must drive under strong interactions with mainline traffic [2,3]. As illustrated in Fig. 1, the EV is required to generate feasible control commands within a limited merging space and enter the mainline traffic at an appropriate time, completing the merging maneuver safely, smoothly, and in compliance with traffic rules while respecting vehicle dynamics [4]. Consequently, on-ramp merging is not only essential for efficient traffic flow but also plays a critical role in ensuring roadway safety and driving comfort [5].

Conventional autonomous merging motion planning approaches are typically based on sampling, search, or optimization. Representative methods include lattice-based sampling [6], artificial potential fields [7], and model predictive control (MPC) [8]. These approaches can be implemented in a hierarchical framework, for example by first generating a merging trajectory through geometric curves [9] and then tracking it via low-level controllers such as PID [10] or sliding-mode control (SMC) [11], etc. Alternatively, trajectory planning and tracking control can be formulated and solved in a unified optimization framework, among which MPC-based methods have been extensively studied [12]. While these methods have demonstrated promising performance in relatively simple scenarios, their applicability to more congested and complex traffic environments remains limited, mainly

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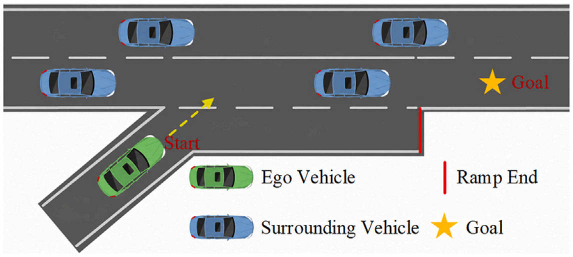


Fig. 1. Autonomous on-ramp merging scenario.

due to reliance on rule-based solvers, manual modeling assumptions and parameter adjustments [13].

To promote high-level autonomous driving in complex and open environments, learning-based approaches, particularly imitation learning (IL) [14] and reinforcement learning (RL) [15], have been extensively studied in recent years. IL is able to directly learn a mapping from state observations to action outputs by leveraging expert driving demonstration data, thereby encoding human driving patterns into the learned policy. In contrast, RL does not rely on expert data and instead enables an agent to learn optimal driving policies through trial-and-error interactions with the environment using reward feedback. Owing to this self-improving mechanism, RL demonstrates stronger scenario adaptability, and has achieved performance surpassing human experts in a variety of sequential decision-making and control tasks, such as the game of Go [16], StarCraft [17], autonomous racing [18], and drone competitions [19].

Nevertheless, the deployment of RL algorithms in highly interactive on-ramp merging tasks still faces several critical limitations: (i) high-dimensional state–action spaces and a large policy exploration domain result in substantial exploration challenges. Moreover, the task is typically driven by sparse terminal rewards associated with successful merging, which further decreases sample efficiency, making it difficult for RL agents to rapidly converge to high-quality policies [20]; (ii) Autonomous merging is a safety-critical scenario in which even occasional unreasonable actions produced by a learned policy may lead to serious collision risks. Purely data-driven RL policies lack explicit safety awareness and often fail to provide sufficient guarantees during policy execution [21]; (iii) Conversely, overly conservative safety guarantees may excessively restrict the agent's action output, degrade driving efficiency and prevent the vehicle from completing merging maneuvers in a timely and effective manner under dense and competitive traffic conditions. [22].

In order to address the aforementioned limitations and further enhance the performance of the RL-based autonomous on-ramp merging approach, this paper proposes a **P**hysics-informed **R**esidual RL framework for safe and efficient autonomous on-ramp merging, in which **E**xpert **P**rior **K**nowledge (**PR-EPK**) is explicitly incorporated to guide both policy learning and execution (as shown in Fig. 2). Specifically, model-based control knowledge derived from an MPC expert is incorporated into policy learning through residual action parameterization, thereby reducing unnecessary exploration and improving sample efficiency in highly dynamic and interactive merging scenarios. In addition, analytical interaction-risk indicators are incorporated into constrained policy learning and risk-adaptive execution. The main contributions of this work are summarized as follows:

1. A physics-informed residual RL framework method is proposed, in which an MPC expert provides control prior knowledge to RL policy learning. By explicitly decomposing the control action into an expert reference and a learnable residual action, this framework significantly reduces useless exploration, improves sample efficiency, and enhances convergence stability in highly interactive merging tasks.
2. We develop a risk-adaptive hybrid policy execution mechanism that integrates an analytical interaction risk assessment module with an MPC-based safety-oriented fallback, enabling adaptive action refinement under high-risk conditions.
3. We introduce a risk-aware constrained policy learning scheme that incorporates risk-sensitive costs, fallback-intervention and risk-adaptive residual regularization. A decoupled task/safety value estimation design and a progressive curriculum strategy are further adopted to improve the safety–efficiency balance and convergence robustness.

This paper is organized as follows: Section 2 introduces the related works on RL autonomous driving. Section 3 introduces the proposed methodology, including the problem preliminaries, the residual RL framework with MPC prior, the risk-adaptive hybrid policy execution and the risk-aware dual-critic policy learning with a two-stage curriculum. Section 4 introduces the experimental setup and results. Finally, Section 5 concludes this work.

2. Related work

In recent years, RL has demonstrated strong adaptability in autonomous driving tasks [23–25]. It learns closed-loop driving policies directly from interaction data, significantly reducing reliance on manually engineered rules [26], and generalizes well to complex and highly interactive conditions. For instance, early studies adopted DQN-style discrete decision-making to select high-level maneuvers in highway scenarios, showing that learned policies can match or surpass commonly used rule-based baselines across different driving tasks [27]. With the need for finer-grained control and closer coupling to motion

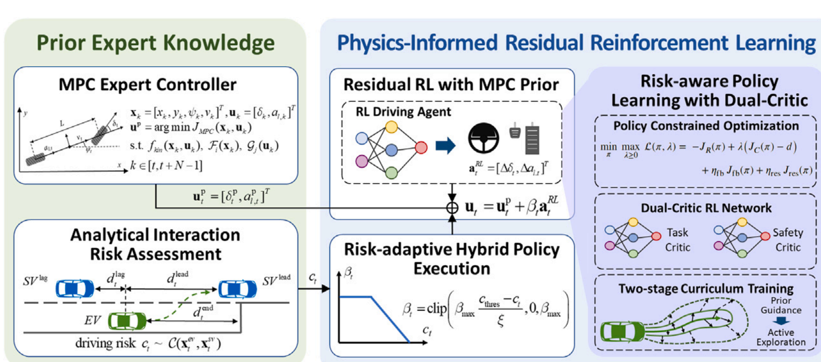


Fig. 2. The MPC expert controller and the analytical risk assessment module provide structured prior knowledge for residual policy learning and execution. Specifically, the MPC generates a dynamically feasible nominal control command, while the RL agent learns a risk-adaptive residual action to refine and improve the expert control.

planning, recent research has increasingly shifted to continuous-control actor–critic methods. SAC has been applied to joint decision-making and motion planning in interactive traffic, explicitly balancing safety, efficiency, and comfort under driving uncertainties [28]. Beyond flat policy learning, hierarchical RL (HRL) further decomposes driving into goal generation and goal-conditioned execution, improving scalability in complex environments, which could be further augmented with state-wise constraints to better enforce state safety during both learning and deployment [29].

However, the effectiveness of RL-based autonomous driving methods is fundamentally challenged by the high-dimensional, continuous-control, and highly dynamic nature of interactive traffic environments [30,31]. In autonomous merging, the large state–action space and sparse terminal rewards lead to low sample efficiency [32]. Moreover, an exploration mechanism driven by a single composite reward may induce oscillatory behaviors near critical safety margins [33], limiting the quality and robustness of the learned driving policy.

To alleviate these limitations, prior expert information has been injected into the RL agent to guide or constrain learning [34]. The most common approaches incorporate prior information either from the perspective of expert data or expert knowledge. Expert-data-driven approaches leverage demonstrations to guide policy learning, typically through imitation learning or demonstration-regularized RL [14]. While such methods can accelerate convergence, they heavily depend on demonstration quality and coverage, and are prone to covariate shift in interactive out-of-distribution states. In contrast, expert-knowledge-driven approaches incorporate structured prior knowledge, such as analytical models, control laws, or safety constraints, into the learning and execution loop [5,35,36]. These priors may serve as nominal references in reward shaping, loss terms in policy optimization, or safety filtering mechanisms that constrain or project policy outputs. Nevertheless, rigid prior designs and mismatched conservatism may introduce authority conflicts or excessive constraint bias. In addition, performance often becomes sensitive to cost limits and may degrade under overly conservative settings [37], leading to brittle behavior or efficiency loss in dense, tight-gap merging scenarios [38,39].

Despite these advances, achieving accelerated learning through physics-grounded expert knowledge while simultaneously ensuring explicit step-wise feasibility and hard safety margins—without inducing undue conservatism in dense, highly interactive traffic—remains an open challenge [39].

A closely related branch of expert-knowledge-guided RL is residual reinforcement learning, where a nominal controller generates a baseline action and the learned policy predicts only a corrective residual. This formulation has been shown to improve sample efficiency and execution robustness in robotic control by combining conventional feedback control with learned residual corrections [40]. Similar residual policy learning frameworks further demonstrate that learning residual actions on top of imperfect hand-designed or MPC controllers can solve long-horizon and sparse-reward manipulation tasks more efficiently than learning full policies from scratch [41]. Beyond action-level residual correction, MPC has also been blended with learned value-function approximation to balance model bias and value-estimation errors during policy learning [42]. Recent studies have further extended residual MPC/RL frameworks to real-world social navigation and online policy customization, where prior policies or controllers are adapted during execution through residual learning or online planning [43,44].

Nevertheless, most existing residual RL studies focus on robotic manipulation, generic continuous control, social navigation, or execution-time policy adaptation, rather than traffic-structured interaction risks in autonomous merging. PR–EPK addresses this gap by embedding MPC-based residual learning into a merging framework, where the MPC prior is jointly integrated with risk-adaptive execution, safety fallback, dual-critic constrained learning, and curriculum-based authority release. This enables safer and more efficient residual policy learning under dense interactive traffic.

3. Methodology

3.1. Problem preliminaries

3.1.1. Vehicle model

The EV is modeled with a discrete-time kinematic bicycle model. The state vector is defined as $\mathbf{x}_t = [x_t,\ y_t,\ \psi_t,\ v_t]^\top$, where (x_t, y_t) denotes the rear-axle center position, ψ_t is the heading angle, and v_t is the longitudinal velocity. The control input vector consists of $\mathbf{u}_t = [\delta_t,\ a_{l,t}]^\top$, where δ_t is the front-wheel steering angle and $a_{l,t}$ is the longitudinal acceleration. Then the vehicle kinematic model can be formulated as:

$$\mathbf{x}_{t+1} = f(\mathbf{x}_t, \mathbf{u}_t) = \begin{bmatrix} x_t + v_t \cos(\psi_t)\Delta t \\ y_t + v_t \sin(\psi_t)\Delta t \\ \psi_t + \dfrac{v_t}{L}\tan(\delta_t)\Delta t \\ v_t + a_{l,t}\Delta t \end{bmatrix}, \tag{1}$$

where L denotes the wheelbase, and to account for actuation limitations, the control inputs are limited to $\delta_t \in [-\delta_{\max}, \delta_{\max}]$, $a_{l,t} \in [a_{l,\min}, a_{l,\max}]$.

This kinematic model forms the necessary foundation for incorporating prior vehicle control knowledge into the RL framework.

3.1.2. Markov decision process

The on-ramp merging problem can be formulated as a sequential decision-making problem, which can be constructed as a Markov Decision Process (MDP) via the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma, \rho_0 \rangle$, where $\mathcal{S}$ and $\mathcal{A}$ denote the observation state space and action space of the agent, respectively. The transition $\mathcal{P}(s_{t+1}|s_t, a_t)$ encapsulates the probability of the agent transitioning to the next state. $r(s_t, a_t)$ represents the reward when the agent takes action a_t in the state s_t, $\gamma \in (0, 1)$ is the discount factor characterizing the temporal urgency and ρ_0 is the initial state distribution. A policy $\pi : \mathcal{S} \to \mathcal{P}(\mathcal{A})$ maps given states to action distributions. The standard RL objective is to maximize the expected discounted return $J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} \gamma^t r(s_t, a_t)\right]$ with a policy $\pi(a_t|s_t) \in \Pi$ (Π denotes the set of all policies) and T denotes the rollout horizon used for return estimation rather than an online planning horizon.

For safety-critical problems, reward maximization alone may lead the RL agent to unsafe actions. Therefore, the standard MDP problem could be further extended to a constrained MDP (CMDP)[45], i.e., to maximize the expected return while satisfying the cost constraints, d_i is the constraint limit:

$$\begin{aligned} &\max_\pi J(\pi) \quad \text{s.t. } \pi \in \Pi_C = \{\pi \in \Pi | \forall i, J_{C_i}(\pi) < d_i\}. \\ &\quad J_{C_i}(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} \gamma^t c_t\right] \end{aligned} \tag{2}$$

3.1.3. RL-based autonomous merging

In on-ramp merging scenarios, the EV must merge into the target lane within limited space while strongly interacting with surrounding vehicles (SVs). The objective is to learn a closed-loop motion control policy that generates feasible continuous control commands to identify an appropriate merge gap and safely complete lane insertion within a time/space limit. A merging attempt is considered unsuccessful if a collision occurs, the EV leaves the drivable region, or the time limit is exceeded.

Under the CMDP formulation above, the autonomous merging problem is specified by the following state and action definitions. The state $s_t \in \mathcal{S}$ compactly encodes the EV motion state and the surrounding traffic context:

$$s_t \triangleq \left[\mathbf{x}_t^{ev},\ \mathbf{e}_t,\ d_t^{\text{end}},\ \{\mathbf{o}_t^i\}_{i \in \mathcal{N}_t}\right], \tag{3}$$

where $\mathbf{x}_t^{ev} = [x_t,\ y_t,\ \psi_t,\ v_t]^\top$ denotes the EV pose and speed, $\mathbf{e}_t$ denotes the tracking errors with respect to the target lane centerline, d_t^{end} denotes the remaining longitudinal distance to the ramp-end reference station, and $\mathbf{o}_t^i$ denotes the relative state of the selected surrounding vehicle SV_i with respect to the EV.

Rather than directly outputting full control commands, the RL policy produces a residual action

$$a_t \triangleq \mathbf{a}_t^{\mathrm{RL}} = \Delta \mathbf{u}_t^{\mathrm{RL}} = [\Delta\delta_t,\ \Delta a_{l,t}]^\top \in \mathcal{A}, \tag{4}$$

where $\Delta\delta_t$ and $\Delta a_{l,t}$ denote the residual steering and longitudinal-acceleration adjustments, respectively. The detailed integration of this residual action with the MPC prior is described in the following subsection.

The environment evolves according to the coupled vehicle–traffic dynamics, inducing state transitions $s_{t+1} \sim \mathcal{P}(\cdot \mid s_t, a_t)$. Therefore, autonomous merging is formulated as a CMDP-based motion planning problem, in which the policy $\pi(a_t \mid s_t)$ is optimized to maximize the expected return while satisfying safety-related cost constraints.

3.2. Residual RL framework with MPC prior

To incorporate reliable model-based control knowledge while retaining the flexibility of RL, we develop an MPC-based residual RL framework in which a receding-horizon MPC controller based on vehicle kinematics provides dynamically feasible nominal control inputs, while the RL agent learns bounded residual actions to refine these commands. By operating in the residual action space around the MPC-defined baseline, the RL agent focuses on learning driving task-relevant corrections rather than full control actions, which significantly reduces low-value random exploration, improves sample efficiency, and enhances policy stability.

3.2.1. Residual action realization

Following the action definition in (4), the policy output is applied as a residual correction to the MPC prior command for online control execution.

The executed continuous control input is defined as $\mathbf{u}_t \triangleq [\delta_t,\ a_{l,t}]^\top$, where δ_t is the steering angle and $a_{l,t}$ is the longitudinal acceleration. Both variables are normalized from their physical bounds to $[0, 1]$ for policy learning.

To improve control feasibility and reduce inefficient exploration, the executed control is decomposed into an MPC prior command $\mathbf{u}_t^{\mathrm{p}} \triangleq [\delta_t^{\mathrm{p}},\ a_{l,t}^{\mathrm{p}}]^\top$ (provided by our previous work [46]), and the learned residual policy output. The final executed control input can be expressed as:

$$\mathbf{u}_t = \mathbf{u}_t^{\mathrm{p}} + \beta_t\, \mathbf{a}_t^{\mathrm{RL}}, \tag{5}$$

where $\beta_t \in [0, 1]$ is the residual authority factor. Here, the MPC prediction horizon N is strictly internal to the MPC module for computing the nominal command $\mathbf{u}_t^p$.

This residual formulation avoids direct action blending between the MPC and RL outputs, which might otherwise result in ineffective control commands due to inherently competing objectives.

3.2.2. Reward design

The reward function is defined as the sum of a task-oriented component and a safety-related component: $r_t = w_{\mathrm{task}} r_t^{\mathrm{task}} + w_{\mathrm{safe}} r_t^{\mathrm{safe}}$.

The task-related reward is designed to capture multiple aspects of the merging objective, including desired speed tracking, control smoothness during merging, and terminal goal rewards:

$$\begin{aligned}
r_t^{\mathrm{task}} &= r_t^{\mathrm{eff}} + r_t^{\mathrm{com}} + r_t^{\mathrm{goal}},\\
r_t^{\mathrm{eff}} &= w_v[2\,\sigma\Big(k\big(\tfrac{v_t}{v_{\mathrm{des}}} - c\big)\Big) - 1] + w_d(\frac{d_t^{\mathrm{obs}} - d_{\mathrm{crit}}}{d_{\mathrm{safe}} - d_{\mathrm{crit}}}),\\
r_t^{\mathrm{com}} &= 1 - \Big[w_a\,\mathrm{clip}(a_{l,t} - a_{l,t-1})^2 + w_\delta\,\mathrm{clip}(\delta_t - \delta_{t-1})^2\Big],\\
r_t^{\mathrm{goal}} &= \begin{cases} g_{succ}, & \text{successful merge}\\ g_{coll}, & \text{collision}\\ g_{else}, & \text{off-road or frozen}\end{cases}
\end{aligned} \tag{6}$$

Here, w_v and w_d weight the speed tracking and ramp end distance in the efficiency term. v_t is the EV's speed and v_{des} is the desired driving speed. The sigmoid function $\sigma(\cdot)$ parameterized by (k, c) shapes the speed–tracking reward by providing near–neutral feedback around the desired speed range while penalizing large deviations. The distance term evaluates the remaining feasible distance d_t^{obs} using two thresholds $(d_{\mathrm{safe}}, d_{\mathrm{crit}})$, producing a higher score when the EV has more remaining distance and penalizing encroachment into the critical region. The comfort term discourages abrupt acceleration/braking and sharp steering maneuvers through corresponding weights w_a, w_δ. It is formulated as a normalized quadratic penalty on action changes to ensure control smoothness. The terminal goal reward explicitly encodes the outcome of a merging attempt, providing a positive reward for successful completion and penalties for failure cases, including collision, leaving the road boundary, and exceeding the allowed time.

The safety–related reward is constructed from physically interpretable risk metrics, including time–to–collision (TTC) and safe distance between vehicles:

$$\begin{aligned}
r_t^{\mathrm{safe}} &= w_{\mathrm{ttc}}\, r_t^{\mathrm{TTC}} + w_{\mathrm{gap}}\, r_t^{\mathrm{sd}}\\
r_t^{\mathrm{TTC}} &= \begin{cases} +0.05, & \mathrm{TTC}_t \ge \theta,\\ -1.0, & \text{otherwise,}\end{cases}\\
r_t^{\mathrm{sd}} &= \frac{\min\big(d_t^{\mathrm{lead}}, d_t^{\mathrm{lag}}\big) - d_{\mathrm{safe}}(v_t)}{\kappa},
\end{aligned} \tag{7}$$

Specifically, the TTC–based safety reward assigns penalties to situations in which the predicted TTC falls below a predefined threshold θ, reflecting potential collision risk. In addition, the safety distance incorporates the longitudinal distance with both the lead and lag SVs in the target lane (d_t^{lead} and d_t^{lag}). To ensure sufficient speed–adaptive clearance, $d_{\mathrm{safe}}(v_t)$ is formulated using two distinct physical bounds: $d_{\mathrm{safe}}(v_t) = \max\big(d_{\mathrm{space}}(v_t),\ d_{\mathrm{time}}(v_t)\big)$. Here, the spatial bound $d_{\mathrm{space}}(v_t) = d_{\min} + k_v v_t$ guarantees a fundamental kinematic buffer, while the temporal bound $d_{\mathrm{time}}(v_t) = d_{\min} + v_t T_{\mathrm{gap}}$ ensures a sufficient reaction time headway on the minimum distance, where T_{gap} is a predefined desired time-gap constant. Furthermore, $\kappa > 0$ is a distance normalization constant for the spacing deficit term, representing the characteristic scale of gap insufficiency. It is set as a fixed constant based on the typical spacing range in the merging scenario. Together, these metrics provide complementary safety cues that account for collision risk and interaction feasibility in merging scenarios.

3.3. Risk–adaptive hybrid policy execution

Purely reward–driven RL policies may produce locally aggressive actions that violate safety margins in on–ramp merging scenarios. To ensure deployment–phase reliability, (i) we design a risk–adaptive hybrid policy execution mechanism. It is able to project geometrickinematic observations into a compact risk index; (ii) performs risk–adaptive blending and MPC takeover with hysteresis to avoid chattering; and (iii) reuses the same risk signal to shape risk–aware policy learning, improving the intrinsic safety quality of the learned actions and reducing intervention frequency.

3.3.1. Analytical interaction risk assessment

To provide explicit risk-aware guidance during learning and execution, the EV motion states are projected onto the target lane to establish a unified longitudinal reference for evaluating merging feasibility and vehicle-interaction risk. Based on the projected longitudinal states, a fused risk cost $c_t \in [0, 1]$ (higher means higher risk) is constructed as:

$$\begin{cases} c_t^{\text{ttc}} = \sigma\left(\dfrac{\tau_{\text{ttc}} - TTC_t}{\kappa_{\text{ttc}}}\right), \\ c_t^{\text{sd}} = \min\left(1, \dfrac{[d_{\text{safe}}(v_t) - s_t]_+}{\Delta s}\right), \\ c_t^{\text{end}} = \min\left(1, \left[1 - \dfrac{d_t^{\text{end}}}{D_{\text{ramp}}}\right]_+\right), \end{cases} \tag{8}$$

where c_t^{ttc} is a TTC-based risk term that maps TTC_t to $[0, 1]$ and increases as the time-to-collision decreases; $(\tau_{\text{ttc}}, \kappa_{\text{ttc}})$ control the transition location and smoothness, respectively. c_t^{sd} evaluates whether the available longitudinal gap s_t under the projected target-lane reference is sufficient at speed v_t relative to the speed-adaptive safe distance $d_{\text{safe}}(v_t)$, where Δs denotes the deficiency range before saturation and $[x]_+ = \max(x, 0)$ is the positive-part operator. Accordingly, c_t^{sd} is zero when the available gap is sufficient, increases linearly as the gap becomes insufficient, and saturates at one under severe spacing deficiency. c_t^{end} is a ramp-end risk defined by the remaining distance to the ramp terminal d_t^{end}, normalized by the ramp length D_{ramp}, and increases monotonically as the EV approaches the ramp end. Finally, the fused risk score is conservatively aggregated as $c_t = \max\left(c_t^{\text{ttc}}, c_t^{\text{sd}}, c_t^{\text{end}}\right)$, allowing the most critical risk source to govern the overall risk estimate used in constrained policy learning and risk-adaptive execution.

3.3.2. Residual authority adaptation

Within the residual action space defined in Eq. (5), a risk–adapted hybrid policy execution mechanism is introduced which continuously attenuates residual RL action influence as the agent approaches a safety–critical regime.

Let $\xi > 0$ denote a prescribed risk margin that defines the attenuation range around the activation threshold c_{thres}. Residual authority is scheduled by the risk cost c_t as

$$\beta_t = \text{clip}\left(\beta_{\max}\,\frac{c_{\text{thres}} - c_t}{\xi},\ 0,\ \beta_{\max}\right), \tag{9}$$

so that β_t smoothly decreases to zero with increased c_t, yielding a continuous reduction of residual intervention near the safety boundary.

The MPC–based safe control fallback is activated whenever the risk cost violates the admissible envelope ($c_t > c_{\text{thres}}$) or an emergency TTC guard is triggered ($TTC_t < \tau_{\text{take}}$). Under such conditions, the residual action a_t^{RL} influence is suppressed by setting $\beta_t = 0$, and the executed control command degenerates to the constraint–satisfying MPC solution. To prevent undesirable chattering behavior near the switching boundary, a lightweight hysteresis mechanism is introduced. Specifically, once the fallback mode is activated, it remains engaged for at least a minimum dwell time H. The system is allowed to return to residual execution only when stricter recovery conditions are satisfied, i.e., $c_t < c_{\text{rel}}$ and $TTC_t > \tau_{\text{rel}}$, where $c_{\text{rel}} < c_{\text{thres}}$ and $\tau_{\text{rel}} > \tau_{\text{take}}$. This design ensures smooth and stable switching while preserving real–time implementability.

3.4. Risk–aware policy learning with dual–critic

3.4.1. Policy constrained optimization

We incorporate risk awareness directly into the constrained policy learning process, such that safety considerations are reflected not only during execution but also during policy optimization. Specifically, we leverage the evaluated risk index to construct risk–sensitive constraint costs, enabling the RL agent to proactively maintain safety margins while learning effective merging behaviors.

In addition, we introduce two independent regularization terms into the optimization objective: (i) an intervention regularizer and (ii) a risk–adaptive residual regularizer. The former penalizes excessive reliance on MPC–based expert control fallback, encouraging the policy to maintain safety margins without recurrent external intervention. The latter suppresses aggressive residual corrections in near–critical regimes, promoting smoother and more stable control behavior near safety boundaries. We optimize a Lagrangian–regularized objective:

$$\max_{\pi}\ \min_{\lambda \geq 0}\ \mathcal{L}(\pi, \lambda) = J_R(\pi) - \lambda\left(J_C(\pi) - d\right) - \eta_{\text{fb}}\, J_{\text{fb}}(\pi) - \eta_{\text{res}}\, J_{\text{res}}(\pi), \tag{10}$$

The objective components are defined as:

$$\begin{aligned} J_R(\pi) &= \mathbb{E}_\pi\left[\sum_{t=0}^{T-1} \gamma^t r_t\right], J_C(\pi) = \mathbb{E}_\pi\left[\sum_{t=0}^{T-1} \gamma^t c_t\right], \\ J_{\text{fb}}(\pi) &= \mathbb{E}_\pi\left[\sum_{t=0}^{T-1} \gamma^t m_t\right], J_{\text{res}}(\pi) = \mathbb{E}_\pi\left[\sum_{t=0}^{T-1} \gamma^t c_t\ \left\|\mathbf{a}_t^{\text{RL}}\right\|_2^2\right], \end{aligned} \tag{11}$$

where J_R is the expected discounted return, $J_C(\pi)$ represents the cumulative safety cost under the risk constraint with an allowable safety threshold d. $J_{\text{fb}}(\pi)$ and $J_{\text{res}}(\pi)$ are two additional regularization terms. Specifically, $J_{\text{fb}}(\pi)$ quantifies the expected discounted frequency of MPC fallback activations, where $m_t \in 0, 1$ is a binary indicator: $m_t = 1$ indicates that the expert MPC controller fallback is used, and $m_t = 0$ otherwise. This term discourages excessive reliance on fallback intervention and promotes sustained agent's autonomous behavior. $J_{\text{res}}(\pi)$ penalizes large residual corrections in near–critical regimes. The Lagrange multiplier $\lambda \geq 0$ adaptively enforces the safety constraint during training, while η_{fb} and η_{res} balance autonomy preservation and residual smoothness within the feasible safety region.

3.4.2. Dual–critic RL network design

Within the proposed PR–EPK framework, the training process is built on a dual–critic network and optimized using a Proximal Policy Optimization (PPO)-style clipped update [47]. Given on–policy rollouts, advantages are estimated via Generalized Advantage Estimation (GAE):

$$\hat{A}_t = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l \delta_{t+l},\ \ \delta_t = r_t + \gamma V(s_{t+1}) - V(s_t), \tag{12}$$

We introduce two independent critic networks to estimate the state values induced by task-oriented and safety-oriented reward components, respectively, thereby decoupling heterogeneous value estimation to accommodate both driving safety and efficiency. The task critic $V_t(s_t; \omega_t)$ estimates the return induced by the task reward r_t^{task}, while the safety critic $V_s(s_t; \omega_s)$ estimates the return induced by the safety reward r_t^{safe}. The overall policy advantage is formed as a linear combination: $\hat{A}_t^{\text{R}} = w_{\text{task}} \hat{A}_t^t + w_{\text{safe}} \hat{A}_t^s$, where $\hat{A}_t^t$ and $\hat{A}_t^s$ are estimated using $V_t(s_t; \omega_t)$ and $V_s(s_t; \omega_s)$, respectively. This formulation can be viewed as a dual-critic implementation of a single weighted composite reward $r_t^{\text{R}} = w_{\text{task}} r_t^{\text{task}} + w_{\text{safe}} r_t^{\text{safe}}$, for which the corresponding value function satisfies $V_t^R = w_{\text{task}} V_t(s_t; \omega_t) + w_{\text{safe}} V_s(s_t; \omega_s)$ by the linearity of discounted returns and expectations. Therefore, under fixed weights and identical discount/GAE settings, the proposed linear combination only provides a scalar advantage estimate for PPO-Clip; the actor is still updated using the same probability ratio and clipping operator. Thus, the dual-critic design does not alter the PPO-Clip objective structure, but reorganizes value estimation for heterogeneous task and safety rewards.

The instantaneous cost c_t, fallback indicator m_t, and residual penalty $c_t \|\mathbf{a}_t^{\text{RL}}\|_2^2$ are computed analytically at each step. Based on Eq.(10), we estimate their discounted returns directly from rollouts:

$$\begin{aligned} \hat{G}_t^C &= \sum_{l=0}^{T-t-1} \gamma_c^{\,l}\, c_{t+l},\ \hat{G}_t^{\text{fb}} = \sum_{l=0}^{T-t-1} \gamma^{\,l}\, m_{t+l}, \\ \hat{G}_t^{\text{res}} &= \sum_{l=0}^{T-t-1} \gamma^{\,l}\left(c_{t+l}\, \|\mathbf{a}_{t+l}^{\text{RL}}\|_2^2\right). \end{aligned} \tag{13}$$

Then, the total policy update signal can be expressed as: $\hat{A}_t^{\text{tot}} = \hat{A}_t^R - \lambda\, \hat{G}_t^C - \eta_{\text{fb}}\, \hat{G}_t^{\text{fb}} - \eta_{\text{res}}\, \hat{G}_t^{\text{res}}$. The actor network is updated by maximizing the PPO clipped surrogate objective:

$$\mathcal{L}_{\text{actor}}(\theta) = \mathbb{E}\Big[\min\Big(\rho_\theta \hat{A}_t^{\text{tot}},\ \text{clip}(\rho_\theta, 1-\epsilon, 1+\epsilon)\hat{A}_t^{\text{tot}}\Big)\Big], \qquad \rho_\theta = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t) \tag{14}$$

The two critics are trained independently via bootstrapped regression:

$$\hat{R}_t^t = \hat{A}_t^t + V_t(s_t;\omega_t),\ \mathcal{L}_{V_t}(\omega_t) = \mathbb{E}\Big[\big(V_t(s_t;\omega_t) - \hat{R}_t^t\big)^2\Big], \qquad \hat{R}_t^s = \hat{A}_t^s + V_s(s_t;\omega_s),\ \mathcal{L}_{V_s}(\omega_s) = \mathbb{E}\Big[\big(V_s(s_t;\omega_s) - \hat{R}_t^s\big)^2\Big]. \tag{15}$$

Since the task and safety rewards enter the learning objective in an additive form, their corresponding value estimates can be decomposed and recombined according to the linearity of expectation. The resulting weighted advantage remains a valid scalar estimate for the PPO-style clipped update. Under function approximation, separate critics are mainly beneficial when the two returns differ in scale, sparsity, or temporal structure, or yield poorly aligned critic-regression gradients, because decoupling can reduce value-estimation interference. Therefore, the dual-critic design preserves the standard actor update form while reorganizing value estimation for heterogeneous task and safety objectives.

3.4.3. Two–stage curriculum training

To stabilize training under highly interactive merging dynamics, we adopt a two–stage curriculum that couples interaction difficulty with a stage–wise release of residual authority. Specifically, we anneal the authority factor β_t with the global interaction step n using a piecewise schedule:

$$\beta(n) = \begin{cases} \rho\,\beta_{\max}\,\dfrac{1-\zeta^{\,n}}{1-\zeta^{\,n_{\text{sw}}}}, & n \le n_{\text{sw}}, \\ \beta_{\max} - (1-\rho)\beta_{\max}\,\zeta^{\,(n-n_{\text{sw}})}, & n > n_{\text{sw}}, \end{cases} \tag{16}$$

where $\zeta \in (0,1)$ is an exponential annealing base, n_{sw} denotes the stage–switch interaction step from Stage 1 to Stage 2, and $\rho \in (0,1)$ specifies the fraction of the maximum residual authority released by the end of Stage 1. The parameters n_{sw} and ζ are fixed curriculum hyperparameters selected from preliminary validation runs to balance training stability and policy autonomy, rather than online risk-triggered variables. Therefore, Stage 1 operates under restricted authority, while Stage 2 further releases authority toward $\beta_{\max}$.

Stage 1: Expert–guided exploration ($\mathcal{D}_1$). Under low–to–moderate traffic density, we intentionally restrict residual authority $\beta(n)$ and emphasize expert guidance to improve sample efficiency and accelerate convergence. By starting from simplified interaction settings, the agent quickly acquires safe and dynamically feasible merging behaviors with stabilized exploration.

Stage 2: Autonomous policy enhancement ($\mathcal{D}_2$). After establishing stable baseline performance, we increase interaction difficulty and progressively release residual authority. This enables the RL agent to further explore beyond expert priors, refine merging strategies, and achieve improved performance under dense and tightly coupled traffic conditions.

The complete training procedure of the proposed PR–EPK method is summarized in Algorithm 1. The detailed policy update process is illustrated in Fig. 3.

Algorithm 1 Risk–Aware Multi–Critic PPO with MPC Prior.

Initialize: curriculum $\{\mathcal{D}_k\}$; MPC prior π_p; actor π_θ; critics $V_t(\omega_t), V_s(\omega_s)$; actor network learning rate α_π; critic network learning rate α_t, α_s; discount γ; rollout T; epochs E; residual schedule $(\beta_{\max}, c_{\text{thres}}, \xi)$; weights $(\omega_t, \omega_s, \alpha_\lambda)$; cost limit d; Lagrange $\lambda \leftarrow \lambda_0$ (cap $\lambda_{\max}$); buffer $\mathcal{B} \leftarrow \emptyset$.
for curriculum stage $k = 1, 2, \ldots$ **do**
  Set env $\mathcal{E} \sim \mathcal{D}_k$.
  for each episode **do**
    $\mathcal{B} \leftarrow \emptyset$.
    for $t = 0, \ldots, T-1$ **do**
      $u_t^p \leftarrow \pi_p(s_t)$; $a_t^{RL} \leftarrow \pi_\theta(\cdot|s_t)$
      Calculate c_t by Eq.(8)
      $\beta_t \leftarrow \text{clip}\Big(\beta_{\max}\,\frac{c_{\text{thres}}-c_t}{\xi},\ 0,\ \beta_{\max}\Big)$
      $u_t = u_t^p + \beta_t a_t^{RL}$
    end for $\mathcal{L}_{\text{V}_\text{t}}$ and $\mathcal{L}_{\text{V}_\text{s}}$.
    Calculate loss functions $\mathcal{L}_{\text{actor}}$, $\mathcal{L}_{\text{V}_\text{t}}$ and $\mathcal{L}_{\text{V}_\text{s}}$ by Eqs. (14) and (15).
    Update actor network
    $\theta \leftarrow \theta - \alpha_\pi \nabla_\theta \mathcal{L}_{\text{actor}}(\theta)$.
    Update critic networks
    $\omega_t \leftarrow \omega_t - \alpha_t \nabla_{\omega_t} \mathcal{L}_{V_t}(\omega_t)$
    $\omega_s \leftarrow \omega_s - \alpha_s \nabla_{\omega_s} \mathcal{L}_{V_s}(\omega_s)$
    Update lagrange multiplier
    $\hat{J}_C \leftarrow \frac{1}{|\mathcal{B}|}\sum_{t\in\mathcal{B}} \hat{G}_t^C$
    $\lambda \leftarrow \text{clip}\big(\lambda + \alpha_\lambda(\hat{J}_C - d),\ 0,\ \lambda_{\max}\big)$
  end for
end for

4. Experiment implementation

4.1. Experiment setting

The training environment is constructed based on the open source simulator Highway–env[48]. The on–ramp merge scenario consists of two mainline lanes and one ramp lane. A uniform lane width is set as $W_{lane} = 4.0\,\text{m}$, and vehicle dimensions are standardized to $L_v = 4.8\,\text{m}$ and $W_v = 1.8\,\text{m}$. The simulator runs at a simulation frequency of 10 Hz, while the policy frequency is set to 2 Hz, and the generated command is held over the intermediate simulation steps.

Traffic density is controlled by the volume to capacity ratio, which parameterizes the stochastic generation of SVs. In our experiments, we consider three density levels: low, medium, and high traffic, corresponding to 12, 16, and 20 SVs in the mainline scenario, respectively. SVs' behaviors are governed by the MOBIL/IDM model [49]. Episodes terminate upon successful merging, collision, or reaching the 500-step limit. The MPC prior is computed by solving the receding-horizon optimization problem using the IPOPT solver. All experiments, including policy training, testing, and runtime profiling, are conducted on a workstation equipped with an Intel i7-14700 CPU and an NVIDIA GeForce RTX 4060 GPU. The actor and the two critic networks adopt identical multilayer perceptron (MLP) architectures, each comprising three fully connected hidden layers with 256 units and ReLU activations. Detailed hyperparameters are listed in Table 1.

4.2. Comparison models and evaluation metrics

To evaluate the proposed PR–EPK method, we compare it with the following representative baselines:

- DQN [50]: A discrete–control RL baseline that discretizes steering and acceleration, representing value–based decision–making in merging tasks.
- PPO and SAC [47,51]: Standard continuous–control actor–critic methods that directly output continuous steering and acceleration commands.
- PPO-L [52]: A Lagrangian-based constrained PPO baseline that augments the PPO objective with an adaptive safety-cost penalty, enabling continuous-control policy learning while explicitly considering CMDP constraints.

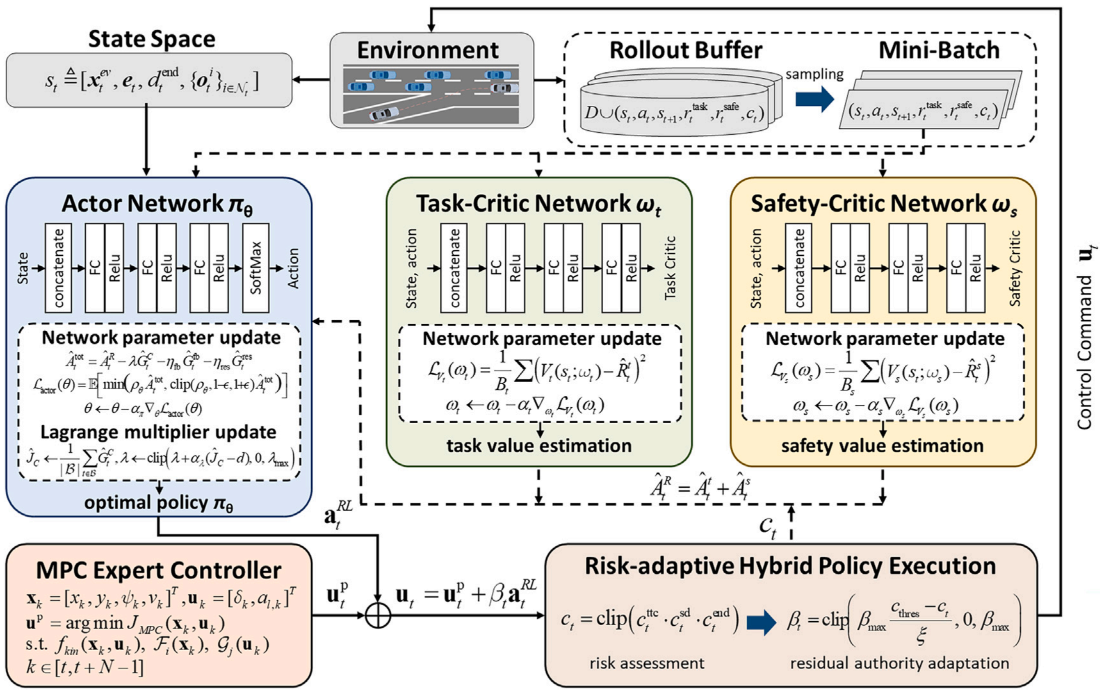


Fig. 3. The policy update process of PR–EPK primarily involves the actor network, task critic and safety critic network, as well as the MPC expert controller and risk–adaptive hybrid policy execution mechanism. The solid line indicates the process of the RL agent interacting with the environment to collect driving experience, while the dashed line represents the parameter updating.

Table 1
Hyper–parameters of RL implementation.

Parameter	Value
Learning rate	$2e-4$
Clip range ϵ	0.20
Target KL δ_{KL}	0.03
Actor network learning rate α_π	$8e-5$
Critic network learning rate α_t/α_s	$8e-5/8e-5$
Discount factor γ	0.97
Mini–batch size B	256
Training steps	5e5
Risk-aware residual attenuation ξ	0.20
Activation threshold c_{thres}	0.35
MPC fallback regularization η_{fb}	0.04
Residual-action regularization η_{res}	0.05
Curriculum switch step n_{sw}	2e5
Minimum dwell time H	$1.0s$
Emergency TTC threshold τ_{take}	$1.8s$
Recovery risk threshold c_{rel}	0.25
Recovery TTC threshold τ_{rel}	$2.5s$
Task reward weight w_{task}	1.0
Safety reward weight w_{safe}	0.5

- PG–SAC[53]: A physics–guided RL that incorporates expert references into the learning objective via a guidance regularization term.
- PR–EPK (Ours): A physics–informed residual RL with a decoupled dual–critic architecture and risk–adaptive hybrid execution mechanism.

In this work, the goal of the EV is to safely and efficiently complete lane merging within the ramp–mainline merging conflict zone, while meeting feasibility constraints. To comprehensively evaluate the proposed methods and baselines, we use the following five metrics:

- **Success Rate (SR):** The fraction of episodes in which the EV reaches the target point safely within the time limit.
- **Collision Rate (CR):** The fraction of episodes that end with a collision or an off–road event.
- **Frozen Rate (FR):** The fraction of episodes in which the EV neither collides nor reaches the target within the time limit, reflecting overly conservative or deadlocked behavior that degrades traffic efficiency. It is computed as FR = 1 − CR − SR.
- **Average Episode Reward (AER):** The mean cumulative reward per episode, measuring overall policy performance during training and evaluation.
- **Average Merge Time (AMT):** The mean elapsed time to complete the merge, computed over successful episodes only; it directly quantifies merging efficiency.

4.3. Comparison results

4.3.1. Training result

Each method was trained with five random seeds, and we report the mean performance with standard–deviation bands. All agents were trained under the medium traffic–density setting.

As shown in Fig. 4(a), DQN performs the worst, converging slowly and achieving the lowest final reward due to its discretized action representation, which limits control fidelity under continuous merging dynamics. PPO and SAC improve upon DQN but still exhibit slower convergence and noticeable oscillations, reflecting their sensitivity to interaction shifts. PPO–L further improves upon vanilla PPO by introducing an adaptive Lagrangian safety–cost penalty, indicating that explicit CMDP constraint awareness benefits continuous–control policy learning. However, its final reward and convergence stability remain below PG–SAC and PR–EPK, since penalty–based constrained learning still learns the control policy largely from scratch and may introduce conservative policy updates. PG–SAC further enhances learning stability and final performance by incorporating MPC–based guidance into policy optimization. In comparison, PR–EPK shows the most significant improvement, with the fastest performance increase during early training and convergence to the highest final reward with the smallest variance. This indicates superior learning efficiency and convergence

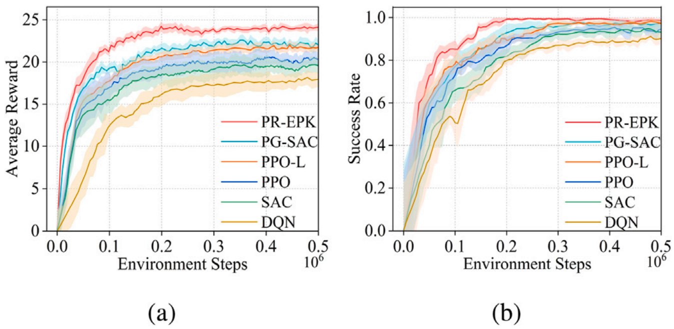


Fig. 4. (a) The average reward of different RL agents during the training process. (b) The success rate of different RL agents during the training process.

robustness compared with all baselines, mainly because the MPC–prior residual parameterization constrains exploration around dynamically feasible controls.

Fig. 4(b) further shows consistent trends in success rate evolution. DQN yields the lowest success rate, followed by PPO and SAC, which plateau at moderate levels with larger fluctuations. PPO–L achieves a higher success rate than vanilla PPO, confirming the benefit of incorporating safety constraints during training. PG–SAC improves both convergence speed and final performance through expert-guided optimization. PR–EPK achieves the highest and most stable success rate, demonstrating strong training stability and reliability in interactive merging scenarios.

4.3.2. Test result

We test all methods under three traffic densities (low, medium, and high) with identical road geometry and randomized background–traffic seeds. For each method and each density, the trained policies from five random seeds are evaluated over 1000 episodes per seed. Fig. 5 presents the results across the three densities: the left column shows the distribution of success rates, while the right column reports the distribution of average merge times over successful episodes. Table 2 further summarizes the corresponding mean statistics with standard deviations.

Across all densities, PPO–L improves over PPO in safety-related metrics by introducing an adaptive Lagrangian safety-cost penalty, confirming that explicit CMDP constraint awareness is beneficial for continuous-control merging tasks. However, PPO–L still remains inferior to PG–SAC and PR–EPK in terms of both safety and efficiency, because penalty-based constrained learning mainly regulates the optimization objective but does not provide a dynamically feasible action prior or a structured risk-aware execution mechanism. PG–SAC consistently outperforms vanilla SAC and PPO in both success rate and merge time, demonstrating the benefit of incorporating physics-grounded expert guidance into RL training. By anchoring policy updates to dynamically feasible control references, PG–SAC achieves improved stability and reduced performance variance compared with purely model-free baselines.

PR–EPK further improves both safety and efficiency by jointly integrating residual policy learning, MPC-based expert control, and risk-aware execution. This design suppresses unsafe residual amplification in tight interactions while preserving policy flexibility. As shown in Table 2, PR–EPK achieves the best overall performance across all densities, consistently yielding the lowest collision rate, highest success rate, and shortest merge time. Although all methods degrade as traffic density increases due to stronger interaction coupling and reduced feasible gaps, PR–EPK exhibits milder degradation and maintains stable performance even in high-density scenarios, demonstrating a robust safety–efficiency balance across varying traffic conditions.

To further evaluate generalization under different SV interaction styles, we vary the parameters of the SV behavior model in the

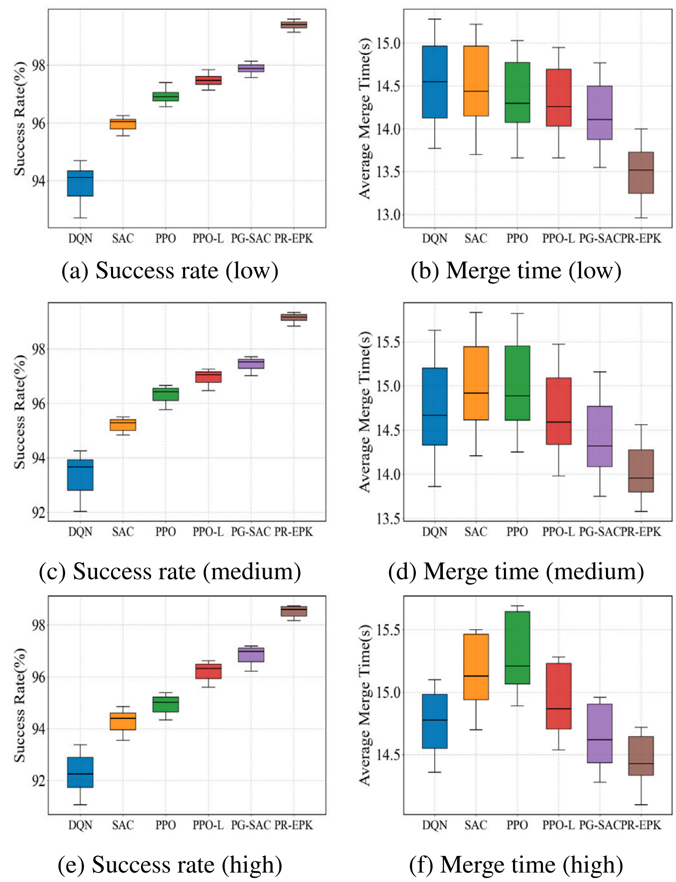


Fig. 5. Testing results under different traffic densities. Each row corresponds to a traffic density (low/medium/high). Left: success rate over All episodes. Right: average merge time over successful episodes.

medium-density scenario. Conservative and aggressive traffic are generated by randomizing the desired time headway and desired jam distance around different nominal centers. For conservative traffic, the sampling centers are set to $T_w = 2.0$ s and $D_w = 12.8$ m, producing larger lead–lag gaps. For aggressive traffic, they are set to $T_w = 1.0$ s and $D_w = 7.8$ m, resulting in tighter gaps and stronger merge conflicts. The normal SV behavior case corresponds to the original medium-density test setting.

As shown in Table 3, all methods degrade under aggressive SV behavior due to reduced feasible merge gaps and stronger interaction conflicts. Nevertheless, PR–EPK maintains the best overall performance, achieving 98.34% SR and 0.54% CR under aggressive traffic. This indicates that PR–EPK generalizes well across different SV behavior patterns, benefiting from the dynamically feasible MPC prior and the risk-aware execution module.

To quantify the online computational overhead, we profile the runtime of PR–EPK over 1000 test episodes under the medium-density scenario. Since episode lengths vary, the latency is measured per policy-update step rather than per episode. As shown in Table 4, the MPC prior computation and RL actor inference take 11.60 ms and 0.42 ms on average, respectively, resulting in a total overhead of 12.02 ms per policy update. The maximum latency is 17.46 ms, still far below the 0.5 s policy-update interval, indicating lightweight online computation.

4.3.3. Robustness evaluation under observation noise

To evaluate robustness against noisy observations, zero-mean Gaussian noise is added to the observation states in the medium-density merging scenario. Two noise levels are considered. In the mild-noise setting, the standard deviations of longitudinal position, lateral position, velocity, and heading angle are set to $\{0.2\text{ m}, 0.05\text{ m}, 0.1\text{ m/s}, 0.5^\circ\}$,

Table 2
Compare performance on three Highway-env test scenarios(mean ± std).

Traffic Density	Models	SR(%)↑	CR(%)↓	FR(%)↓	AER↑	AMT(s)↓
low	DQN	94.12 ± 0.65	5.56 ± 0.65	**0.32 ± 0.18**	21.13 ± 0.78	14.55 ± 0.18
	SAC	96.06 ± 0.54	2.08 ± 0.35	1.86 ± 0.28	23.01 ± 0.62	14.44 ± 0.16
	PPO	96.92 ± 0.48	1.72 ± 0.30	1.36 ± 0.24	23.52 ± 0.56	14.30 ± 0.14
	PPO–L	97.48 ± 0.44	1.00 ± 0.26	1.52 ± 0.21	23.66 ± 0.52	14.26 ± 0.13
	PG–SAC	97.90 ± 0.40	1.12 ± 0.22	0.98 ± 0.18	23.78 ± 0.48	14.11 ± 0.12
	PR–EPK	**99.42 ± 0.21**	**0.18 ± 0.12**	0.40 ± 0.10	**24.89 ± 0.33**	**13.52 ± 0.09**
medium	DQN	93.68 ± 0.78	5.88 ± 0.72	**0.44 ± 0.22**	20.89 ± 0.86	14.67 ± 0.22
	SAC	95.30 ± 0.59	2.20 ± 0.40	2.50 ± 0.32	22.64 ± 0.68	14.92 ± 0.18
	PPO	96.44 ± 0.53	1.98 ± 0.36	1.58 ± 0.27	22.76 ± 0.62	14.89 ± 0.17
	PPO–L	97.06 ± 0.47	1.31 ± 0.30	1.63 ± 0.23	22.82 ± 0.56	14.59 ± 0.15
	PG–SAC	97.54 ± 0.43	1.43 ± 0.26	1.02 ± 0.20	22.94 ± 0.51	14.32 ± 0.13
	PR–EPK	**99.18 ± 0.25**	**0.22 ± 0.14**	0.60 ± 0.22	**24.54 ± 0.38**	**13.96 ± 0.10**
high	DQN	92.26 ± 1.18	6.68 ± 0.88	1.06 ± 0.28	20.52 ± 0.98	14.78 ± 0.26
	SAC	94.40 ± 0.82	2.38 ± 0.48	3.22 ± 0.41	22.40 ± 0.76	15.13 ± 0.23
	PPO	95.02 ± 0.76	3.02 ± 0.56	1.96 ± 0.39	22.32 ± 0.70	15.21 ± 0.22
	PPO–L	96.32 ± 0.63	1.86 ± 0.44	1.82 ± 0.34	22.55 ± 0.63	14.87 ± 0.19
	PG–SAC	96.98 ± 0.56	1.94 ± 0.36	1.08 ± 0.29	22.68 ± 0.58	14.62 ± 0.16
	PR–EPK	**98.60 ± 0.34**	**0.46 ± 0.21**	**0.94 ± 0.27**	**24.22 ± 0.44**	**14.43 ± 0.12**

Table 3
Performance comparison under different surrounding-vehicle behavior styles (mean ± std).

IDM Style	Models	SR(%)↑	CR(%)↓	FR(%)↓	AER↑	AMT(s)↓
Conservative	DQN	94.26 ± 0.72	5.22 ± 0.66	0.52 ± 0.25	21.24 ± 0.82	14.49 ± 0.21
	SAC	95.92 ± 0.57	1.98 ± 0.42	2.10 ± 0.30	22.94 ± 0.66	14.66 ± 0.18
	PPO	96.96 ± 0.52	1.76 ± 0.38	1.28 ± 0.27	23.06 ± 0.61	14.61 ± 0.17
	PPO–L	97.56 ± 0.47	1.16 ± 0.33	1.28 ± 0.24	23.21 ± 0.56	14.36 ± 0.15
	PG–SAC	98.02 ± 0.43	1.22 ± 0.30	0.76 ± 0.21	23.34 ± 0.52	14.15 ± 0.14
	PR–EPK	**99.30 ± 0.26**	**0.20 ± 0.13**	**0.50 ± 0.15**	**24.73 ± 0.37**	**13.72 ± 0.09**
Normal	DQN	93.68 ± 0.78	5.88 ± 0.72	**0.44 ± 0.22**	20.89 ± 0.86	14.67 ± 0.22
	SAC	95.30 ± 0.59	2.20 ± 0.40	2.50 ± 0.32	22.64 ± 0.68	14.92 ± 0.18
	PPO	96.44 ± 0.53	1.98 ± 0.36	1.58 ± 0.27	22.76 ± 0.62	14.89 ± 0.17
	PPO–L	97.06 ± 0.47	1.31 ± 0.30	1.63 ± 0.23	22.82 ± 0.56	14.59 ± 0.15
	PG–SAC	97.54 ± 0.43	1.43 ± 0.26	1.02 ± 0.20	22.94 ± 0.51	14.32 ± 0.13
	PR–EPK	**99.18 ± 0.25**	**0.22 ± 0.14**	0.60 ± 0.22	**24.54 ± 0.38**	**13.96 ± 0.10**
Aggressive	DQN	91.76 ± 1.02	7.30 ± 0.93	**0.94 ± 0.38**	19.93 ± 1.03	15.19 ± 0.30
	SAC	94.18 ± 0.79	2.92 ± 0.60	2.90 ± 0.44	21.84 ± 0.82	15.41 ± 0.24
	PPO	95.08 ± 0.74	2.70 ± 0.55	2.22 ± 0.41	21.96 ± 0.77	15.37 ± 0.23
	PPO–L	95.84 ± 0.67	2.06 ± 0.50	2.10 ± 0.37	22.11 ± 0.72	15.08 ± 0.21
	PG–SAC	96.30 ± 0.61	2.18 ± 0.45	1.52 ± 0.33	22.27 ± 0.67	14.81 ± 0.20
	PR–EPK	**98.34 ± 0.39**	**0.54 ± 0.27**	1.12 ± 0.26	**23.77 ± 0.49**	**14.39 ± 0.15**

Table 4
Online computational time of proposed PR–EPK over 1000 evaluation episodes.

Module	Min (ms)	Max (ms)	Mean (ms)
MPC prior computation	9.94	16.87	11.60
RL actor inference	0.33	0.71	0.42
Total computational time	10.28	17.46	12.02

respectively. In the severe-noise setting, they are increased to $\{1.0\ \mathrm{m}, 0.30\ \mathrm{m}, 0.5\ \mathrm{m/s}, 2.0^\circ\}$. The noise-free case corresponds to the original medium-density test setting.

As shown in Table 5, observation noise degrades the performance of all compared methods, while PR–EPK consistently maintains the best overall performance. Under severe observation noise, PR–EPK still achieves 97.60% SR and 0.74% CR. This degradation is expected because noisy observations directly affect policy inference, MPC prior computation, and risk estimation. Nevertheless, the proposed method remains robust under observation perturbations, benefiting from the dynamically feasible MPC prior and the risk-aware execution mechanism.

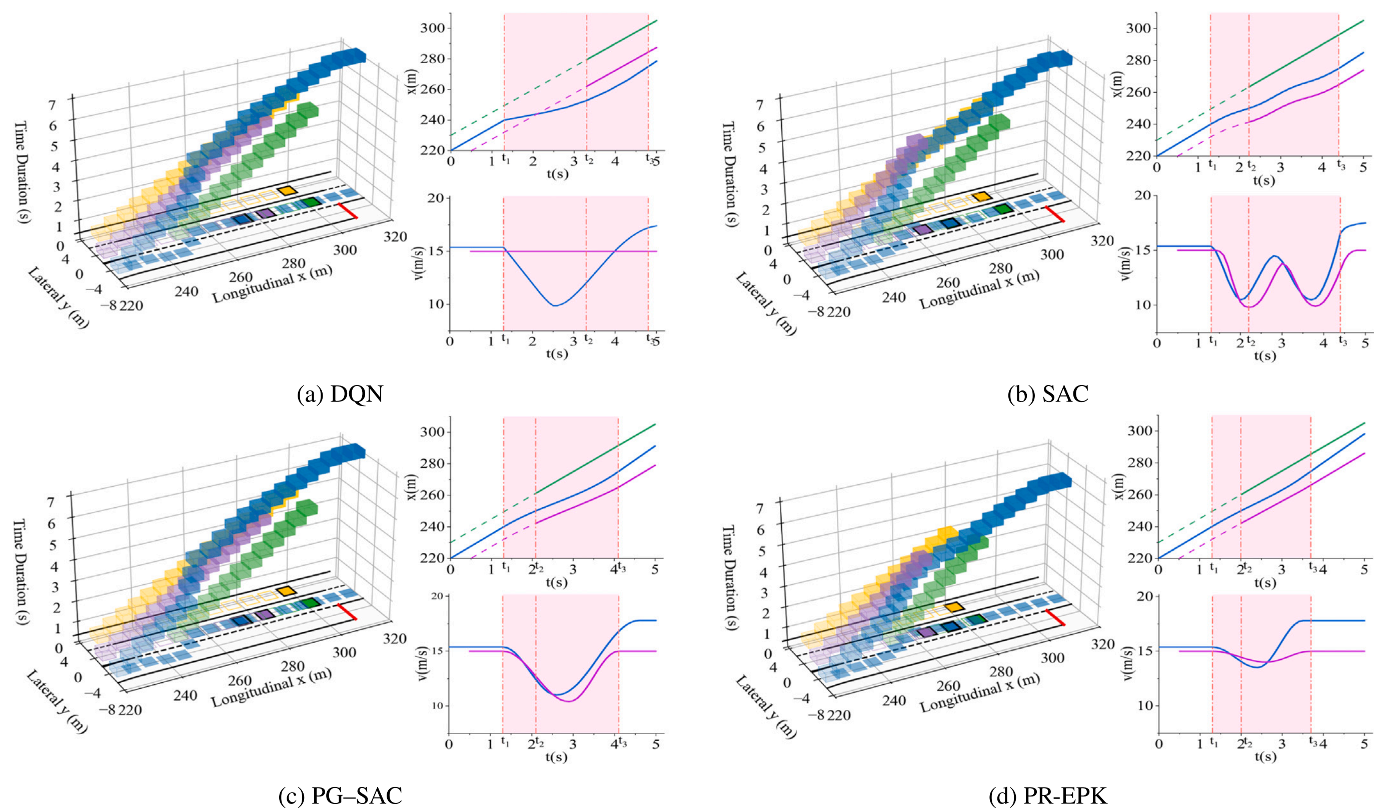
4.3.4. Hyperparameter sensitivity analysis

To examine whether PR–EPK depends on a narrowly manually tuned setting, we conduct a local sensitivity analysis under medium traffic density. We focus on four representative parameters that directly affect the safety–efficiency trade-off: the risk-aware residual attenuation scale ξ, the MPC fallback regularization weight η_{fb}, the residual-action regularization weight η_{res}, and the safety reward weight w_{safe}. The nominal parameters used in the standard medium-density evaluation are $\xi = 0.20$, $\eta_{fb} = 0.04$, $\eta_{res} = 0.05$, $w_{\text{task}} = 1.0$, and $w_{\text{safe}} = 0.5$. For each parameter, we vary only the parameter under investigation while keeping all other parameters fixed, and evaluate the converged policy over 1000 episodes.

As shown in Fig. 7, these parameters mainly regulate the safety–efficiency trade-off by changing residual authority, fallback reliance, residual-action magnitude, and the relative emphasis on task completion and safety preservation. A smaller ξ narrows the residual-attenuation interval and preserves greater residual authority until the risk approaches the activation threshold, improving gap exploitation but increasing risk exposure; a larger ξ attenuates residual actions over a broader risk range, improving safety but increasing conservative or stalled behaviors. A smaller η_{fb} imposes a weaker penalty on MPC fallback, allowing the policy to rely more heavily on the safety safeguard and thereby reducing collisions, whereas a larger η_{fb} promotes more autonomous execution

Table 5
Performance comparison under different observation-noise levels (mean ± std).

Noise	Models	SR(%)↑	CR(%)↓	FR(%)↓	AER↑	AMT(s)↓
None	DQN	93.68 ± 0.78	5.88 ± 0.72	**0.44 ± 0.22**	20.89 ± 0.86	14.67 ± 0.22
	SAC	95.30 ± 0.59	2.20 ± 0.40	2.50 ± 0.32	22.64 ± 0.68	14.92 ± 0.18
	PPO	96.44 ± 0.53	1.98 ± 0.36	1.58 ± 0.27	22.76 ± 0.62	14.89 ± 0.17
	PPO–L	97.06 ± 0.47	1.31 ± 0.30	1.63 ± 0.23	22.82 ± 0.56	14.59 ± 0.15
	PG–SAC	97.54 ± 0.43	1.43 ± 0.26	1.02 ± 0.20	22.94 ± 0.51	14.32 ± 0.13
	PR–EPK	**99.18 ± 0.25**	**0.22 ± 0.14**	0.60 ± 0.22	**24.54 ± 0.38**	**13.96 ± 0.10**
Mild	DQN	92.96 ± 0.88	6.52 ± 0.82	**0.52 ± 0.29**	20.41 ± 0.95	14.83 ± 0.26
	SAC	94.72 ± 0.71	2.60 ± 0.52	2.68 ± 0.40	22.16 ± 0.77	15.06 ± 0.22
	PPO	95.89 ± 0.66	2.27 ± 0.48	1.84 ± 0.35	22.31 ± 0.73	15.02 ± 0.21
	PPO–L	96.54 ± 0.59	1.64 ± 0.44	1.82 ± 0.32	22.43 ± 0.68	14.76 ± 0.19
	PG–SAC	97.02 ± 0.54	1.78 ± 0.39	1.20 ± 0.28	22.57 ± 0.63	14.51 ± 0.17
	PR–EPK	**98.79 ± 0.33**	**0.32 ± 0.19**	0.89 ± 0.24	**24.17 ± 0.45**	**14.11 ± 0.13**
Severe	DQN	89.94 ± 1.38	8.98 ± 1.21	**1.08 ± 0.46**	18.93 ± 1.26	15.46 ± 0.38
	SAC	92.31 ± 1.02	3.86 ± 0.74	3.83 ± 0.58	20.95 ± 1.02	15.71 ± 0.33
	PPO	93.62 ± 0.96	3.62 ± 0.68	2.76 ± 0.52	21.08 ± 0.97	15.62 ± 0.31
	PPO–L	94.71 ± 0.88	2.87 ± 0.61	2.42 ± 0.47	21.35 ± 0.91	15.33 ± 0.29
	PG–SAC	95.26 ± 0.81	3.01 ± 0.56	1.73 ± 0.42	21.54 ± 0.86	15.02 ± 0.27
	PR–EPK	**97.60 ± 0.52**	**0.74 ± 0.33**	1.66 ± 0.39	**23.26 ± 0.64**	**14.63 ± 0.20**

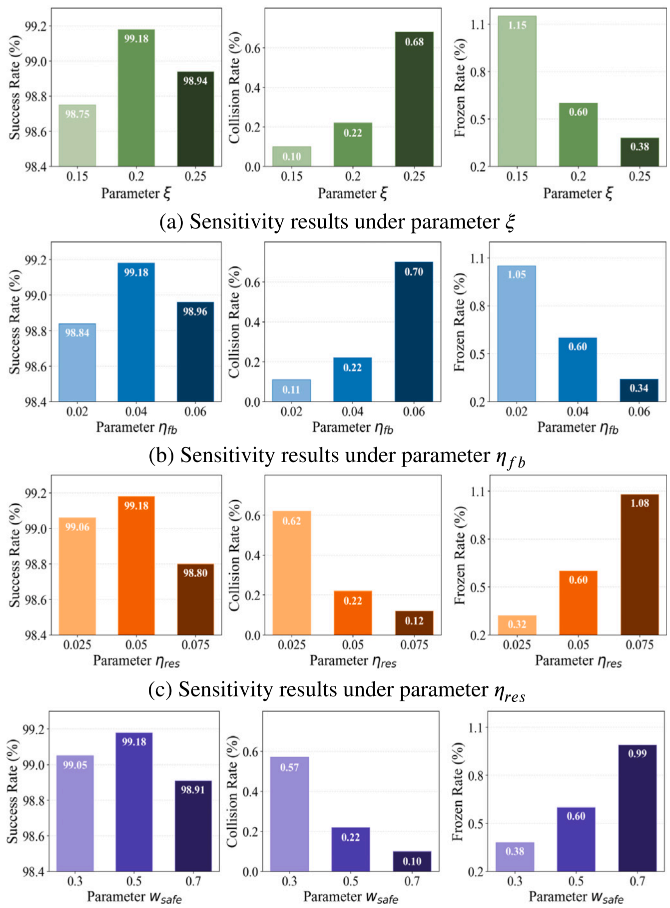


(a) DQN (b) SAC (c) PG–SAC (d) PR-EPK

Fig. 6. Spatiotemporal visualization of EV merging in dynamic traffic: the 3D diagram shows EV/SVs occupancy in longitudinal–lateral–time space, and the right panels show the Corresponding $x - t$ and $v - t$ profiles. In the $x - t$ plot, green/magenta/blue denote the lead vehicle, lag vehicle, and EV trajectories, respectively. t_1, t_2, and t_3 mark lane-change initiation, entry into the target lane, and merge completion. (a) DQN. (b) SAC. (c) PG-SAC. (d) PR-EPK.

but increases collision risk. Similarly, a smaller η_{res} permits stronger residual corrections and improves gap exploitation, while a larger η_{res} constrains the policy to remain closer to the MPC prior, improving safety but increasing conservatism. Finally, a smaller w_{safe} weakens the relative emphasis on safety and encourages more aggressive gap exploitation, reducing frozen behavior at the expense of increased collision risk; a larger w_{safe} strengthens the safety preference and suppresses collisions, but may induce more conservative behavior and reduce task-completion efficiency.

Overall, the sensitivity results show that PR–EPK is not overly dependent on a single narrowly tuned setting within the tested local ranges. Varying ξ, η_{fb}, η_{res}, and w_{safe} mainly shifts the policy along the safety–efficiency spectrum, rather than causing an unstable performance collapse. The nominal setting provides the best overall balance among SR, CR, and FR, reflecting the coordinated effects of risk-aware residual attenuation, MPC fallback regulation, residual-action regularization, and reward-weight balancing.



(a) Sensitivity results under parameter ξ

(b) Sensitivity results under parameter η_{fb}

(c) Sensitivity results under parameter η_{res}

(d) Sensitivity results under parameter w_{safe}

Fig. 7. Parameter sensitivity analysis. Rows represent parameters ξ, η_{fb}, η_{res}, and w_{safe}; columns display success, collision, and frozen rates.

4.3.5. Case study

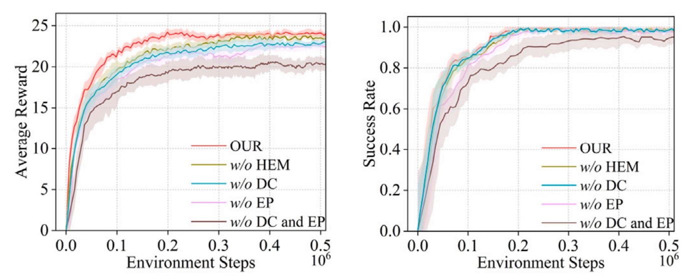
Fig Fig. 6 visualizes the EV merging process using three complementary views: (i) a 3D spatiotemporal occupancy map (longitudinal–lateral–time) that highlights trajectory feasibility and interaction geometry, (ii) an $x-t$ plot where the shaded band denotes the lane-change interval (from lane-change onset to successful insertion), and (iii) a $v-t$ plot that exposes speed coupling between the EV and SVs. For clarity, we only depict the SVs most likely to affect the merge outcome—namely the closest lead (green) and lag (purple) vehicles in the target lane—while other distant vehicles with negligible influence are omitted. In addition, the yellow vehicle represents the nearest vehicle on the adjacent mainline segment around the target-lane neighborhood. In the $x-t$ view, Lead/Lag are rendered as dashed curves before the EV enters the target lane and switch to solid afterward, explicitly indicating the successful insertion.

DQN fails to exploit the early feasible gap and eventually forces the EV to merge behind the lag SV due to the learned policy's hesitation. This behavior is reflected in the delayed lane-change completion and the reactive deceleration in $v-t$. SAC exhibits pronounced oscillations in both trajectory and velocity. The high-frequency fluctuations in the EV's $v-t$ curve propagate to the rear SV, indicating unstable driving behavior and degraded ride comfort. PG-SAC benefits from MPC physical guidance, producing a smoother spatiotemporal corridor and speed adjustment. However, its merging behavior remains conservative, as evidenced by the larger temporal gap with the lead SV maintained before lane insertion. In contrast, PR-EPK delivers the most desirable behavior. It exploits the early feasible gap and completes merging behavior with smooth deceleration and efficient timing. The EV exhibits only mild speed reduction during insertion, and the lag SV experiences only slight velocity variation during the maneuver.

4.4. Ablation study

We ablate the main architecture of PR-EPK that enables *physics-informed residual learning* and *risk-adaptive execution*: (1) full **PR-EPK**; (2) without **HEM** (hybrid execution mechanism), which retains the MPC-guided residual policy but disables risk-adaptive authority regulation and MPC fallback, thereby evaluating the benefit of online risk-aware intervention over fixed residual execution; (3) without **DC** (dual critic), which replaces the decoupled task and safety critics with a standard single critic, directly evaluating whether decoupled value estimation provides benefits over a conventional single-critic implementation; (4) without **EP** (MPC expert prior), which removes MPC guidance and, consequently, the HEM that structurally depends on it, thereby assessing the overall contribution of MPC-based expert knowledge to feasible exploration and safe execution; (5) without **DC** and **EP**, which further removes decoupled value estimation and reduces the framework to pure RL, evaluating the combined contribution of expert guidance and dual-critic learning over a model-free baseline; and (6) pure **MPC**, serving as a model-based baseline without RL. The training curves are shown in Fig. 8, and the test performance under low/medium/high densities is summarized in Table 6.

Overall, full PR-EPK achieves the best training stability and the best overall test performance across all traffic densities, indicating that EP, HEM, and DC contribute complementary rather than redundant benefits.



(a) (b)

Fig. 8. The training results of the ablation study, (a) the average reward of different ablation methods in the training process, (b) the success rate of different ablation methods in the training process.

Table 6
Compare ablation performance on three test scenarios.

Density	Algorithms	SR(%)	CR(%)	FR(%)
low	PR-EPK	99.43	0.18	0.39
	w/o HEM	98.67	0.58	0.75
	w/o DC	98.52	0.66	0.82
	w/o EP	98.33	0.78	0.89
	w/o DC and EP	97.36	1.46	1.18
	MPC	97.88	0.66	1.46
medium	PR-EPK	99.18	0.23	0.59
	w/o HEM	98.35	0.77	0.88
	w/o DC	98.26	0.84	0.90
	w/o EP	98.14	0.92	0.94
	w/o DC and EP	96.86	1.72	1.42
	MPC	97.22	0.89	1.89
high	PR-EPK	98.59	0.47	0.94
	w/o HEM	98.01	0.92	1.07
	w/o DC	97.93	1.01	1.06
	w/o EP	97.78	1.10	1.12
	w/o DC and EP	95.82	2.23	1.95
	MPC	96.18	1.26	2.56

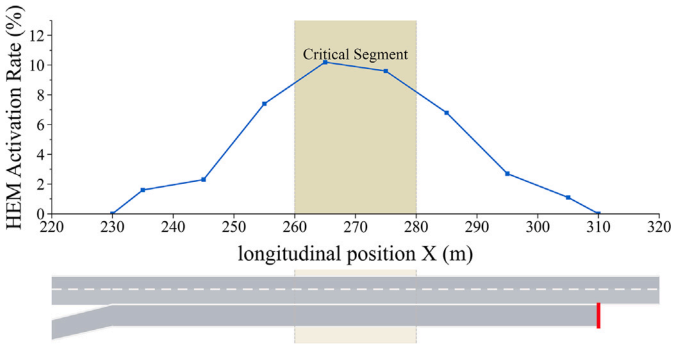


Fig. 9. HEM activation rate along the longitudinal position under the converged PR-EPK policy (high-density, 1000 test episodes). Rates are normalized by state visitation per position bin.

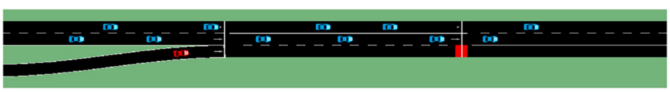


Fig. 10. The on-ramp merging scenario in SUMO.

Here, EP denotes the MPC controller that generates the prior action. Since HEM relies on the MPC fallback branch, it is structurally coupled with EP. Therefore, in the *w/o* HEM variant, EP is retained while risk-adaptive execution is disabled, and a fixed residual-authority is used to preserve the MPC-guided residual control structure.

Removing HEM reduces training stability and degrades risk-related test metrics, indicating that HEM mainly mitigates execution-stage failures in short-horizon critical interactions. It should be regarded as a risk-triggered soft runtime heuristic. This is further supported by Fig. 9, where HEM activations are concentrated in the merge-critical segment and remain rare elsewhere, suggesting that HEM performs sparse risk-aware intervention instead of persistent MPC takeover.

Removing EP leads to slower convergence and lower final performance, showing that the MPC prior primarily mitigates exploration-stage failures by constraining learning around dynamically feasible controls.

Removing DC causes a smaller but consistent degradation. Although the task and safety rewards have comparable per-step magnitudes, the safety signal is sparser and weighted by $w_{\text{safe}} = 0.5$; thus, a single critic trained on the composite return may be dominated by frequent task-related variations and underrepresents safety-critical value changes. The test results support that decoupled value estimation better preserves safety information and improves safety–efficiency coordination.

The pure MPC baseline remains executable but exhibits stronger conservatism, especially in dense traffic, as reflected by increased frozen behavior. Conversely, the *w/o* DC and EP variant, i.e., pure RL, performs worst overall, indicating that dense merging is difficult to solve reliably with either fixed model-based control alone or unconstrained trial-and-error learning alone.

Overall, the ablation results confirm the complementary roles of the three core components: EP guides policy exploration around dynamically feasible nominal controls, HEM provides risk-adaptive execution-stage intervention, and DC stabilizes safety–efficiency learning. Their synergy explains the superior stability and reliability of PR–EPK over both pure MPC and pure RL.

4.5. Cross-simulator validation in SUMO

To evaluate the cross-simulator effectiveness of the proposed method, we further construct an on-ramp merging scenario in the Simulation of Urban Mobility (SUMO) [54], as shown in Fig. 10. The road network consists of a two-lane mainline of about 330 m, an 83.5 m single-lane ramp, and an 84.5 m acceleration lane, with a uniform lane width of 4.0 m. All vehicles are modeled as passenger cars of 4.8 m $\times$ 1.8 m, with a maximum speed of 30 m/s. The EV initial speed is sampled from $[12, 20]$ m/s, while the SV desired-speed factor follows $U(0.6, 1.0)$. The SVs are controlled by the Krauss car-following and LC2013 lane-changing models. Low, medium, and high traffic are generated with per-lane inflows of 600, 1200, and 1800 veh/h, respectively.

For each method and traffic condition, the converged policies trained in Highway-env are directly evaluated in SUMO without additional training or fine-tuning. Specifically, five independently trained policies are tested over 1000 episodes per policy using randomized SUMO traffic seeds. As reported in Table 7, all methods exhibit performance degradation as traffic demand increases, mainly because denser inflows reduce feasible merging gaps and intensify interactions between the EV and SVs. This degradation is particularly evident for conventional model-free baselines. For example, under high traffic demand, DQN suffers from a collision rate of 10.38%, while SAC and PPO show increased frozen or collision rates, indicating that purely data-driven policies are more sensitive to the stronger interaction uncertainty introduced by SUMO traffic dynamics.

The relative performance trends remain consistent with those observed in Highway-env. Under high traffic demand, PPO–L reduces the

Table 7
Compare performance on three test scenarios in SUMO (mean ± std).

Traffic Density	Models	SR(%)↑	CR(%)↓	FR(%)↓	AER↑	AMT(s)↓
low	DQN	91.12 ± 0.88	7.96 ± 0.88	0.92 ± 0.24	20.33 ± 1.05	14.92 ± 0.24
	SAC	94.06 ± 0.73	2.98 ± 0.47	2.96 ± 0.38	22.31 ± 0.84	14.74 ± 0.22
	PPO	95.12 ± 0.65	2.52 ± 0.41	2.36 ± 0.32	22.92 ± 0.76	14.55 ± 0.19
	PPO–L	96.35 ± 0.52	1.35 ± 0.31	2.30 ± 0.25	23.26 ± 0.64	14.43 ± 0.16
	PG–SAC	97.05 ± 0.47	1.45 ± 0.26	1.50 ± 0.20	23.45 ± 0.57	14.23 ± 0.14
	PR–EPK	**99.05 ± 0.23**	**0.30 ± 0.13**	**0.65 ± 0.11**	**24.65 ± 0.38**	**13.58 ± 0.10**
medium	DQN	89.48 ± 1.09	8.98 ± 1.01	1.54 ± 0.31	19.69 ± 1.20	15.22 ± 0.31
	SAC	92.50 ± 0.83	3.50 ± 0.56	4.00 ± 0.45	21.64 ± 0.95	15.42 ± 0.25
	PPO	93.94 ± 0.74	3.08 ± 0.50	2.98 ± 0.38	21.86 ± 0.87	15.34 ± 0.24
	PPO–L	95.35 ± 0.60	1.95 ± 0.38	2.70 ± 0.30	22.18 ± 0.71	14.90 ± 0.19
	PG–SAC	96.15 ± 0.53	1.95 ± 0.32	1.90 ± 0.25	22.42 ± 0.63	14.58 ± 0.16
	PR–EPK	**98.55 ± 0.29**	**0.42 ± 0.16**	**1.03 ± 0.25**	**24.18 ± 0.44**	**14.12 ± 0.11**
high	DQN	87.06 ± 1.71	10.38 ± 1.28	2.56 ± 0.41	18.82 ± 1.42	15.58 ± 0.38
	SAC	90.40 ± 1.19	4.18 ± 0.70	5.42 ± 0.59	21.00 ± 1.10	15.88 ± 0.33
	PPO	91.52 ± 1.10	4.62 ± 0.81	3.86 ± 0.57	21.02 ± 1.02	15.91 ± 0.32
	PPO–L	93.90 ± 0.83	2.85 ± 0.58	3.25 ± 0.44	21.58 ± 0.83	15.39 ± 0.25
	PG–SAC	95.00 ± 0.72	2.75 ± 0.46	2.25 ± 0.36	21.88 ± 0.74	15.03 ± 0.21
	PR–EPK	**97.62 ± 0.40**	**0.78 ± 0.24**	**1.60 ± 0.31**	**23.62 ± 0.53**	**14.72 ± 0.15**

collision rate of PPO from 4.62% to 2.85%, confirming the benefit of explicit safety-cost regulation. PG–SAC further increases the success rate to 95.00% with expert guidance, but its collision rate remains 2.75%. In comparison, PR–EPK achieves the best overall performance, with the highest success rate of 97.62%, the lowest collision rate of 0.78%, and the shortest average merge time of 14.72 s. Compared with PG–SAC, PR–EPK improves the success rate by 2.62% and reduces the collision rate by 1.97%, indicating that its safety improvement is not obtained through overly conservative merging.

Compared with the Highway-env results in Table 2, PR–EPK shows only a moderate performance drop after direct transfer to SUMO. Across low, medium, and high traffic densities, its success rate decreases from 99.42%/99.18%/98.60% to 99.05%/98.55%/97.62%, while the collision rate increases from 0.18%/0.22%/0.46% to 0.30%/0.42%/0.78%. Considering the differences in road geometry, traffic generation, and SV behavior models between Highway-env and SUMO, these results demonstrate the zero-shot cross-simulator effectiveness of the proposed PR–EPK without SUMO-specific retraining.

5. Conclusion

To improve the safety and efficiency of autonomous on-ramp merging, this paper proposes PR–EPK, a physics-informed residual RL framework that integrates structured model-based and analytical expert knowledge into policy learning and execution. An MPC prior provides dynamically feasible nominal controls, while bounded residual learning improves exploration efficiency and policy adaptability. Risk-adaptive execution, decoupled task/safety value estimation, and curriculum-based authority release further enhance safety–efficiency coordination and training stability. Experiments in Highway-env demonstrate that PR–EPK converges faster and consistently outperforms representative baselines in success rate, collision rate, and merge time across traffic densities. Robustness evaluations under observation noise and varied surrounding-vehicle behaviors, together with ablation studies, verify the complementary contributions of the proposed components. Cross-simulator validation in SUMO further shows that PR–EPK retains its performance advantages under different road networks, traffic-generation mechanisms, and vehicle-behavior models, supporting its effectiveness beyond a single simulator. Overall, the results demonstrate that structured expert knowledge can be effectively integrated with residual RL to achieve reliable and efficient autonomous merging. Future work will investigate uncertainty-aware risk modeling, more complex multi-agent scenarios, and sim-to-real transfer.

CRediT authorship contribution statement

Dequan Zeng: Writing – original draft, Software, Methodology. **Zhishao Ni:** Methodology, Data curation, Conceptualization. **Zhuoren Li:** Writing – review & editing, Supervision. **Xinrui Zhang:** Validation, Software. **Ming Liu:** Visualization, Validation. **Yiming Hu:** Software, Conceptualization. **Chen Sun:** Writing – review & editing. **Bo Leng:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available upon request.

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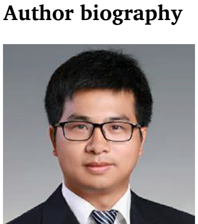
Author biography



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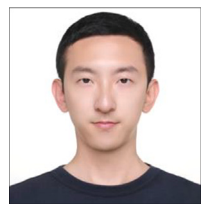

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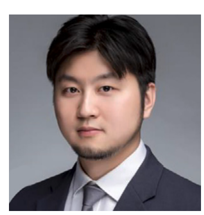

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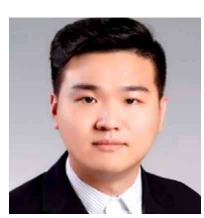

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